

Element C: Presentation and Justification of Solution Design Requirements

Sandiyah Mannah - Design Requirement 1

- **Durability** - it has to last for the rest of the school year
- **Aesthetically pleasing** -
- **Allows student's eyes a break** - can't be too busy or hard to locate since its supposed to help provide students with a visual break
- **Original** -
- **Stands out** - Mr. Gleason has a lot of other decorations in his room so this needs to stand out from those other decoration
- Has to plug into the wall
- **Size** - due to the limited space, it needs to be less than 2 ft by 2 ft
- **Ease of use** - the more complicated it is to use the less it will be used. Therefore it needs to be easy for Mr. Gleason to use and fix if need be.
- **Cost** - we have a limited budget as the engineers so the project can't go above that budget

When doing research for the project, we came across several different fake window designs. Some were simply posters drawn to look like windows while others were complicated devices that looked like windows. After interviewing our stakeholders, we came to understand where these other designs fell short for our usage. For one, most were too big as they were designed to fill up a wall with little room for much else. In addition, they either didn't fulfill the main purpose of an actual window or if they did, they did so for too high of a price. Therefore, our main requirements became: one, it had to function as close to an actual window as possible; two, it couldn't be bigger than 2ft by 2ft; and three, it couldn't cost too much. Other requirements such as ease of use and originality came once those main three requirements were established.

Marilyn Phan - Design Requirement 2

- **Stability** - it has to be able to hold up
- **Size** - it has to fit a specific area of Mr. Gleason's wall
- **Magnetic** - it has to have strong magnets to stick to its magnetic wall
- **Artistically impressive** - it has to look nice, realistic, and represent of nature
- **Light** - it has to have a suitable weight so it won't fall
- **Plug-in feature** - it has to have plugs, batteries cannot be installed into the product
- **Durable** - it has to be able to last long and not be prone to damage in its interior or exterior parts
- **Affordable and resourceful** - it has to be made from resourceful materials to avoid harm to the environment and reduce the price of materials used

These design requirements are important because they focus on the stakeholder's needs. The stakeholder, Mr. Gleason, expressed his necessity for an artificial window in his classroom that depicts nature. Due to the usual dark and gloomy atmosphere in most classrooms, Mr. Gleason wants to encourage and motivate the spirits of his students. As a result, a window consisting of beautiful scenery, outstanding LED lights, and captivating the attention of uninspired or unmotivated individuals would

improve the quality of not only the students but also the subject that is being taught. Since Mr. Gleason's walls are magnetic, the window should be lightweight and have magnetic features allowing it to stick to the wall. In addition, it must have a plug-in feature to power the LED lights, so the stakeholders can make decisions to turn it off or on. The window must also be a reasonable size, considering the teachers' limited wall space.

Vivian Nguyen- Design Requirement 3

- **Aesthetically pleasing** - The fake window has to be nature-themed and somewhat realistic to give off the illusion of an actual window.
- **Lightweight** - Needs to be lightweight for it to stay on the wall, above the floor.
- **Durable** - Needs to be durable for storage and moving purposes.
- **Strong (Using magnets)** - Helps keep the window on the wall, preventing breakage and injuries in case it falls on someone.
- **Long lasting lights** - Helps with the longevity of the product and costs less in the long run when it comes to replacements.
- **Small size (2ft x 2ft)** - For Mr. Gleason, it must be no more than 2ft x 2ft to fit the small space he has for the window.
- **Plug-In (No Batteries)** - Mr. Gleason asked for a plug-in window. No batteries will be used,
- **Cheap** - Needs to be cost effective for what we build and can help save money if replacements are needed.
- **Customizable** - Change the brightness of the light to fit the atmosphere of the room.

These design requirements are specifically tailored to the needs of the stakeholder. With the information from the interview, the requirements are mostly catered to his room. His classroom is mostly magnet walls, explaining why we need magnets to attach the window instead of the regular nail. The size is also rather small to make sure we can fit it within the space he's given us. The adjustable light helps give the window more of a natural brightness. Since Mr. Gleason turns off his room lights often, using yellow light lamps instead; it would be better if he could change the brightness to better fit the overall aesthetic and mood. It can still pop out in the room, while not being too bright to distract the students from working.

An Le- Design Requirement 4

- **Aesthetically pleasing-** The visuals of the project must be pleasing to the eye, if not the project would be taken down. This will be decided by our stakeholders, Mr. Gleason and his students.
- **Magnetic (Detachable)-** For the convenience of Mr. Gleason, the window must be magnetic. Drilling holes into the wall is unacceptable.
- **Plugged in (No battery)-** Mr. Gleason prefers the window to be low maintenance, meaning he does not want to replace batteries. This project must plug into the wall.
- **Light-** The artificial window must light up in order to fully mimic a real window.
- **Lightweight-** To prevent the window from falling and putting less stress on the magnets. The goal is to have the window equal to or less than five pounds.
- **Durability-** The final product must be strong and durable so that the product can be used for years and years.

- **Nature-** To more accurately mimic a real window, the scenery of the window must display nature. This is also recommended by our stakeholder, Mr. Gleason.
- **Size-** To fit into the given space and to listen to the preferences of Mr. Gleason the window must be no bigger than 2ft by 2ft.

A large element of being an engineer is to have engineering empathy. This refers to an engineer's ability to understand the wants and needs of their clients/stakeholders. Without engineering empathy, the final product will fail to cater to the stakeholders. Having aesthetic, magnetic, non-battery operated, 2ft by 2ft, and nature-mimicking requirements for the window will keep the interests of Mr. Gleason, the main stakeholder, in mind and be easier attended to. The additional requirements such as illumination, weight, and durability, are requirements that were added to keep the design engaging and reusable for years to come. According to prior research, having no natural lights in classrooms can make schools feel suffocating. By making the fake window illuminate classrooms will have the illusion of an open space with natural lighting. Making the window lightweight and durable will encourage Mr. Gleason to hang it up every year for his students.